Biopharmaceutical Technology

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Biopharmaceutical Technology (A07850)

Short Title:	Biopharmaceutical Technology		
Faculty	Science and Computing		
Department	Science		
Cluster	Biology		
Author	LCOFFEY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module will focus on the production and function of biopharmaceutical molecules, in particular protein-based and immunotherapeutic drugs.

Programmes

`BIOT-0003`	BSc (Hons) in Applied Biology with Quality Management (`WD_SQMBI_B`)	4 / 2 / M
`BIOL-0029`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	4 / 8 / M

Indicative Content

- Protein production methods and considerations: recombinant and native
- Protein analysis and purification methods, including downstream processing
- Proteins as analytical tools
- Antibody technology towards the production of immunotherapeutic drugs
- Key biopharmaceutical drugs currently available: characteristics and mechanism of action

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Comment on native and recombinant protein expression and purification tasks, including issues relevant to biopharmaceutical practice.
2. Evaluate modern protein engineering, purification, detection and quantification techniques.
3. Appraise the applications of proteins as analytical tools as relevant to the biomedical and biopharmaceutical industries.
4. Comment on the application of antibody technology towards the design and manufacture of immuno-based therapies.
5. Compare modern protein-based biopharmaceutical drugs of choice, including sub-topics such as production, mechanism of action and side-effects.
6. Develop optimised virtual protein purification and analysis protocols using relevant software.

Learning and Teaching Methods

- Lectures.
- Practicals.
- Problem-solving sessions.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5
Continuous Assessment	50%	
Practical	45%	2
Assignment	5%	6

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.

40%-49%: Will be able to show limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.

50%-59%: As well as the above, the student will be able to show reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all topics contained in the learning outcomes and display good competence with regards to use of associated laboratory instrumentation and very good laboratory practice.

70%-100%: All of the above to excellent levels. In addition the learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Tutorial	12	
Practical	36	
Independent Learning	75	

Essential Material

- "Protein virtual lab." University of Leeds website. http://home.btconnect.com/agbooth/archive/swingPP/ProtLab.html
- Subramanian, G. _Biopharmaceutical Production Technology_. Germany: Wiley-VCH , 2012.

Requested Resources

- Room Type: Biology Lab